

Anushka Pradeep Chauhan

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OBJECTIVE

Robotics student specializing in ROS2-based autonomous systems, perception, and SLAM for real-world robotic deployment, Experienced in sensor fusion, robotic manipulation, and computer vision for intelligent robotics and automation systems

EDUCATION

Northeastern University, Boston, MA

Dec 2026

Master of Science in Robotics, Electrical and Computer Engineering

Relevant Courses: Mobile Robotics, Robot Sensing and Navigation, Robotics Mechanics Control, Assistive Robotics, Human robot interaction, Robotics science and systems, Legged Robotics

University of Mumbai, Mumbai, India

May 2024

Bachelor of Engineering in Electronics and Telecommunication

Relevant Courses: Signals & Systems, Electronic Instrumentation & Control Systems, Embedded Systems, Digital System Design

TECHNICAL SKILLS

Programming: Python, C++, MATLAB, Linux, Git, Docker

ROS & Robotics: ROS, ROS2, MoveIt2, TF2, SLAM, Navigation, Motion Planning, Robot Manipulation

Perception: OpenCV, YOLOv8, Vision Transformers, Depth Estimation, LiDAR + RGB-D Sensor Fusion

Simulation & Tools: Gazebo, NVIDIA Isaac Sim, USD Scene Graph, Real-to-Sim Validation

Hardware: RealSense D405, Kinect RGB-D, Raspberry Pi, Embedded Linux Systems

PROJECTS

Autonomous Robotic Manipulator (Stretch RE3), Northeastern University

Feb 2025 - Apr 2025

- Developed autonomous pouring in ROS2 with modularized control & perception nodes, **achieving a 90% success rate in task completion during trials**
- Developed a real-to-sim pipeline using **RealSense D405 data and TF2 transforms**, aligning perception modules with simulated scenes in Gazebo and USD-compatible environments
- Explored NeRF-style RGB-D reconstruction approaches as a proof of concept for scene representation in manipulation
- Validated in Gazebo and hardware** under **embedded Linux** with iterative tuning, reducing response latency by 30%

Robust Incremental Smoothing and Mapping, Northeastern University

Sep 2024 - Dec 2024

- Integrated **robust SLAM** modules to guide **3D reconstruction pipelines** from real world sensor data, evaluated performance for downstream simulation fidelity in USD compatible scenes
- Applied **SIG kernels and trust region optimization** to refine **trajectory estimation**, improving computational efficiency by **15%**
- Validated riSAM** on real world datasets via drift analysis, lowering localization drift by 15% and enabling reliable navigation

Industrial Robotic Arm, University of Mumbai

Sep 2023 - Mar 2024

- Trained **YOLOv8 with Kinect depth data in ROS** for real time package detection, reducing classification errors by **35%**
- Automated** Cartesian XYZ trajectories via **Raspberry Pi and PCA9685 drivers**, improving responsiveness and lowering handling errors by **20%**
- Integrated **sensor feedback** and optimized communication protocols, cutting response times by 30%

Doppler Based Directional Awareness Wearable, Northeastern University

Sep 2025 - Dec 2025

- Developed a **sound localization and motion classification pipeline** using GCC-PHAT + Doppler shift analysis
- Achieved $\pm 10^\circ$ direction accuracy and < 100 ms response in simulation, forming basis for **embodied perception research**
- Designed as a proof-of-concept for assistive auditory systems to enhance spatial awareness in wearable robotics

EXPERIENCE

Tatum Robotics, Boston, MA

Jan 2026 - Present

Hardware & Embedded Systems, Intern

- Defining system requirements for an assistive input device to ensure reliable and responsive human-to-device interaction
- Reviewing embedded input pipelines including GPIO handling, firmware logic, and host-side communication
- Evaluating microcontroller platforms and validation workflows for assistive device prototyping/system testing

K R Petroleum, Mumbai, India

Jun 2022 - Mar 2023

Industrial Automation, Intern

- Developed PLC control logic for pump systems; built Python/SQL pipelines for predictive maintenance (downtime 15%)
- Integrated flow/level sensors with automation logic, improving accuracy by 12% and reducing manual intervention by 40%

PUBLICATION

Patil, R. S., Apraj, A., Bhosale, S., Borikar, S., & Chauhan, A. "Elevating Industrial Picker Arm through Software Mechanic Fusion," IEEE ICNEWS, 2024. [DOI](#)